

卷积优化实验报告

PB25050968 于墨涵 2026.7.23

下列优化版本中，base 为基于 baseline 函数实现的优化基准版本；v1-v5.0 均在 base 基础上分别进行单项优化，用于评估各优化策略的独立效果；v5.1 尝试在 v5.0 基础上进一步优化，但由于代码实现问题未能完成；final 版本则整合了 v1-v5.0 中验证有效的优化策略。

base: 无优化

```
=== Performance Benchmark (Geometric Mean) ===
Kernel size: 5 x 5
Rounds per size: 5

Max absolute error: 0.000000e+00
N= 512 | TPE_Base: 60.64 ns | TPE_Opt: 61.60 ns | Speedup: 0.98x
Max absolute error: 0.000000e+00
N=1024 | TPE_Base: 61.08 ns | TPE_Opt: 61.70 ns | Speedup: 0.99x
Max absolute error: 0.000000e+00
N=2048 | TPE_Base: 68.71 ns | TPE_Opt: 70.18 ns | Speedup: 0.98x
Max absolute error: 0.000000e+00
N=4096 | TPE_Base: 81.34 ns | TPE_Opt: 82.44 ns | Speedup: 0.99x
Max absolute error: 0.000000e+00
N=8192 | TPE_Base: 82.74 ns | TPE_Opt: 108.93 ns | Speedup: 0.76x

Overall Speedup (Geometric Mean): 0.935x
```

v1: 基于提示 2.4 进行优化（将列访问改为行访问，且利用分块技术）

```
=== Performance Benchmark (Geometric Mean) ===
Kernel size: 5 x 5
Rounds per size: 5

Max absolute error: 2.384186e-07
N= 512 | TPE_Base: 60.34 ns | TPE_Opt: 61.57 ns | Speedup: 0.98x
Max absolute error: 2.384186e-07
N=1024 | TPE_Base: 61.38 ns | TPE_Opt: 60.94 ns | Speedup: 1.01x
Max absolute error: 2.384186e-07
N=2048 | TPE_Base: 67.95 ns | TPE_Opt: 61.14 ns | Speedup: 1.11x
Max absolute error: 2.980232e-07
N=4096 | TPE_Base: 82.05 ns | TPE_Opt: 66.91 ns | Speedup: 1.23x
Max absolute error: 2.980232e-07
N=8192 | TPE_Base: 82.83 ns | TPE_Opt: 68.66 ns | Speedup: 1.21x

Overall Speedup (Geometric Mean): 1.102x
```

小数据基本没有优化效果，而对于大尺度数据优化效果较好，与理论一致

v2: 基于提示 2.2 进行优化：引入变量 tmp_sum，先将数据存储如寄存器后写入，

```

=== Performance Benchmark (Geometric Mean) ===
Kernel size: 5 x 5
Rounds per size: 5

Max absolute error: 0.000000e+00
N= 512 | TPE_Base: 61.31 ns | TPE_Opt: 54.62 ns | Speedup: 1.12x
Max absolute error: 0.000000e+00
N=1024 | TPE_Base: 62.52 ns | TPE_Opt: 54.99 ns | Speedup: 1.14x
Max absolute error: 0.000000e+00
N=2048 | TPE_Base: 69.30 ns | TPE_Opt: 62.26 ns | Speedup: 1.11x
Max absolute error: 0.000000e+00
N=4096 | TPE_Base: 82.41 ns | TPE_Opt: 78.07 ns | Speedup: 1.06x
Max absolute error: 0.000000e+00
N=8192 | TPE_Base: 83.33 ns | TPE_Opt: 86.34 ns | Speedup: 0.97x

Overall Speedup (Geometric Mean): 1.077x

```

优化效果一般

v3: 不调用函数

```

=== Performance Benchmark (Geometric Mean) ===
Kernel size: 5 x 5
Rounds per size: 5

Max absolute error: 0.000000e+00
N= 512 | TPE_Base: 60.74 ns | TPE_Opt: 60.97 ns | Speedup: 1.00x
Max absolute error: 0.000000e+00
N=1024 | TPE_Base: 61.18 ns | TPE_Opt: 60.40 ns | Speedup: 1.01x
Max absolute error: 0.000000e+00
N=2048 | TPE_Base: 67.05 ns | TPE_Opt: 65.40 ns | Speedup: 1.03x
Max absolute error: 0.000000e+00
N=4096 | TPE_Base: 82.74 ns | TPE_Opt: 78.50 ns | Speedup: 1.05x
Max absolute error: 0.000000e+00
N=8192 | TPE_Base: 83.05 ns | TPE_Opt: 90.81 ns | Speedup: 0.91x

Overall Speedup (Geometric Mean): 0.999x

```

优化效果并不明显，可能是编译器已经做了 inline 或者调用函数的时间消耗在这个程序中并不大。

v4.1 和 v4.2 均基于 2.1 消除重复计算的方向优化

v4.1: 利用宏替换 KERNEL_RADIUS，避免循环内多次计算

```

=== Performance Benchmark (Geometric Mean) ===
Kernel size: 5 x 5
Rounds per size: 5

Max absolute error: 0.000000e+00
N= 512 | TPE_Base: 60.71 ns | TPE_Opt: 61.75 ns | Speedup: 0.98x
Max absolute error: 0.000000e+00
N=1024 | TPE_Base: 61.16 ns | TPE_Opt: 61.32 ns | Speedup: 1.00x
Max absolute error: 0.000000e+00
N=2048 | TPE_Base: 67.32 ns | TPE_Opt: 67.56 ns | Speedup: 1.00x
Max absolute error: 0.000000e+00
N=4096 | TPE_Base: 82.23 ns | TPE_Opt: 82.44 ns | Speedup: 1.00x
Max absolute error: 0.000000e+00
N=8192 | TPE_Base: 83.67 ns | TPE_Opt: 103.43 ns | Speedup: 0.81x

Overall Speedup (Geometric Mean): 0.954x

```

基本没有任何优化效果，可能是编译器宏替换过程中已经做了优化了

v4.2: 在上级函数中计算 k_size,k_rad; 调用函数时增加参数, 达成循环外提

```
=== Performance Benchmark (Geometric Mean) ===
Kernel size: 5 x 5
Rounds per size: 5

Max absolute error: 0.000000e+00
N= 512 | TPE_Base: 60.57 ns | TPE_Opt: 70.38 ns | Speedup: 0.86x
Max absolute error: 0.000000e+00
N=1024 | TPE_Base: 61.32 ns | TPE_Opt: 70.02 ns | Speedup: 0.88x
Max absolute error: 0.000000e+00
N=2048 | TPE_Base: 67.16 ns | TPE_Opt: 72.61 ns | Speedup: 0.92x
Max absolute error: 0.000000e+00
N=4096 | TPE_Base: 83.05 ns | TPE_Opt: 81.63 ns | Speedup: 1.02x
Max absolute error: 0.000000e+00
N=8192 | TPE_Base: 85.70 ns | TPE_Opt: 90.17 ns | Speedup: 0.95x

Overall Speedup (Geometric Mean): 0.924x
```

小数据量负优化, 大数据量微小提升 (而且似乎只是波动)。也是考虑可能本身编译器已经进行了优化

v5.0: 在图像边界加两层 "0", 避免了循环内层的分支判断

```
=== Performance Benchmark (Geometric Mean) ===
Kernel size: 5 x 5
Rounds per size: 5

Max absolute error: 0.000000e+00
N= 512 | TPE_Base: 60.43 ns | TPE_Opt: 51.09 ns | Speedup: 1.18x
Max absolute error: 0.000000e+00
N=1024 | TPE_Base: 61.04 ns | TPE_Opt: 52.22 ns | Speedup: 1.17x
Max absolute error: 0.000000e+00
N=2048 | TPE_Base: 67.15 ns | TPE_Opt: 54.37 ns | Speedup: 1.23x
Max absolute error: 0.000000e+00
N=4096 | TPE_Base: 81.72 ns | TPE_Opt: 69.90 ns | Speedup: 1.17x
Max absolute error: 0.000000e+00
N=8192 | TPE_Base: 85.87 ns | TPE_Opt: 71.72 ns | Speedup: 1.20x

Overall Speedup (Geometric Mean): 1.190x
```

确实是效果最好的一步优化

v5.1: 尝试在 v5.0 基础上展开卷积循环, 但是一直 bus error 报错, 找不到原因。不过问题应该还是下标判断出错了。

```
=== Performance Benchmark (Geometric Mean) ===
Kernel size: 5 x 5
Rounds per size: 5

make: *** [perf] Bus error: 10
```

思路: 我是把卷积计算当成题干举例那样一列内的 kernel 值相等, 合并计算, 同时在开始时一并计算 10 个偏移值, 减少重复计算。

final: 整合了 v1、v2、v5.0 三个有效优化

```
=== Performance Benchmark (Geometric Mean) ===
Kernel size: 5 x 5
Rounds per size: 5

Max absolute error: 2.384186e-07
N= 512 | TPE_Base: 60.55 ns | TPE_Opt: 47.01 ns | Speedup: 1.29x
Max absolute error: 2.384186e-07
N=1024 | TPE_Base: 62.02 ns | TPE_Opt: 46.97 ns | Speedup: 1.32x
Max absolute error: 2.384186e-07
N=2048 | TPE_Base: 68.53 ns | TPE_Opt: 47.33 ns | Speedup: 1.45x
Max absolute error: 2.980232e-07
N=4096 | TPE_Base: 82.33 ns | TPE_Opt: 61.44 ns | Speedup: 1.34x
Max absolute error: 2.980232e-07
N=8192 | TPE_Base: 82.78 ns | TPE_Opt: 61.66 ns | Speedup: 1.34x

Overall Speedup (Geometric Mean): 1.347x
```

总结：实验内，我除循环展开外实现了所有题干给出的系统层优化方向。其中，除减少函数调用以及消除重复计算优化效果不明显，其他均显著提升了程序运行效率。